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JEE MAIN · QUESTION PAPER

JEE Main 2025 (Online) 29th January Evening Shift

JEE Main Question Paper with Answer Key

29 Jan 2025	75	English
HELD ON	QUESTIONS	LANGUAGE

INSIDE THIS PDF

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Answer key	at the end

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Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement (I): On nitration of m-xylene with HNO_3, H_2SO_4 followed by oxidation, 4-nitrobenzene-1,3-dicarboxylic acid is obtained as the major product.

Statement (II) : $-CH_3$ group is o/p-directing while $-NO_2$ group is m-directing group.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Both Statement I and Statement II are true

(C) Both Statement I and Statement II are false

(D) Statement I is false but Statement II is true

Q2. MCQ (Single Correct Answer)

Open solution

Identify the essential amino acids from below:

(A) Valine

(B) Proline

(C) Lysine

(D) Threonine

(E) Tyrosine

Choose the correct answer from the options given below:

(A) (A), (C) and (D) only

(B) (C), (D) and (E) only

(C) (A), (C) and (E) only

(D) (B), (C) and (E) only

Q3. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement (I): NaCl is added to the ice at 0°C, present in the ice cream box to prevent the melting of ice cream.

Statement (II): On addition of NaCl to ice at 0°C, there is a depression in freezing point.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are true

(D) Statement I is false but Statement II is true

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Q4. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement (I): In partition chromatography, stationary phase is thin film of liquid present in the inert support.

Statement (II): In paper chromatography, the material of paper acts as a stationary phase.

In the light of the above statements, choose the **correct** answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Q5. MCQ (Single Correct Answer)

Open solution

0.1 M solution of KI reacts with excess of H_2SO_4 and KIO_3 solutions. According to equation

$5I^- + IO_3^- + 6H^+ \rightarrow 3I_2 + 3H_2O$

Identify the correct statements :

(A) 200 mL of KI solution reacts with 0.004 mol of KIO_3

(B) 200 mL of KI solution reacts with 0.006 mol of H_2SO_4

(C) 0.5 L of KI solution produced 0.005 mol of I_2

(D) Equivalent weight of KIO_3 is equal to ($\frac{\text{Molecular weight}}{5}$)

Choose the **correct** answer from the options given below :

(A) (C) and (D) only **(B)** (A) and (B) only **(C)** (B) and (C) only **(D)** (A) and (D) only

Q6. MCQ (Single Correct Answer)

Open solution

The type of oxide formed by the element among Li, Na, Be, Mg, B and Al that has the least atomic radius is :

(A) AO **(B)** AO_2 **(C)** A_2O **(D)** A_2O_3

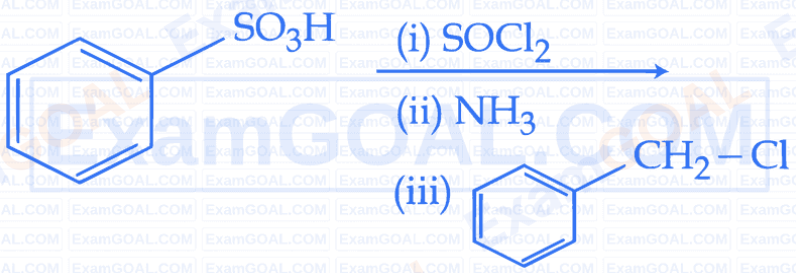
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Q7. MCQ (Single Correct Answer)

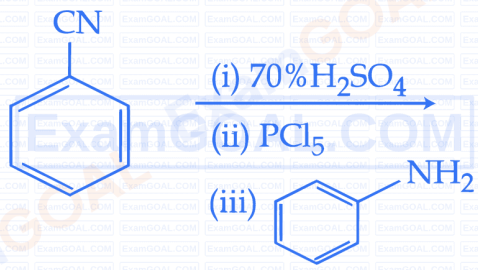
Open solution

Which one of the following reaction sequences will give an azo dye?

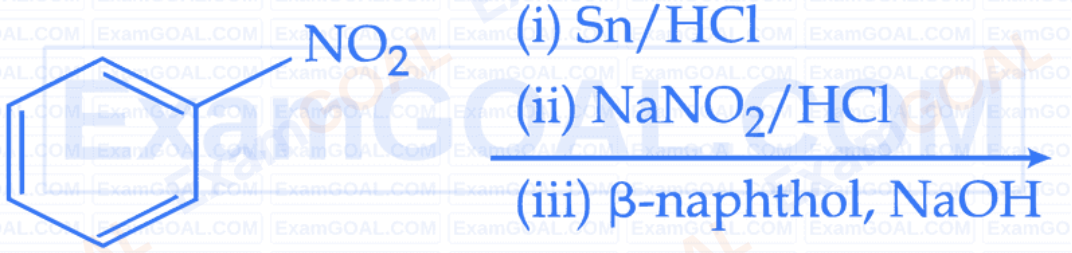
(A)



(B)



(C)



(D)



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Q8. MCQ (Single Correct Answer)

Open solution

Identify the homoleptic complexes with odd number of d electrons in the central metal :

(A) $[FeO_4]^{2-}$

(B) $[Fe(CN)_6]^{3-}$

(C) $[Fe(CN)_5NO]^{2-}$

(D) $[CoCl_4]^{2-}$

(E) $[Co(H_2O)_3F_3]$

Choose the **correct** answer from the options given below :

(A) (C) and (E) only

(B) (A), (B) and (D) only

(C) (B) and (D) only

(D) (A), (C) and (E) only

Q9. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II :

List - I (Applications)	List - II (Batteries/Cell)
(A) Transistors	(I) Anode - Zn/Hg; Cathode - HgO + C
(B) Hearing aids	(II) Hydrogen fuel cell
(C) Inverters	(III) Anode - Zn; Cathode - Carbon
(D) Apollo space ship	(IV) Anode - Pb; Cathode - Pb | PbO_2

Choose the correct answer from the options given below :

(A) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(B) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(C) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(D) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

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Q10. MCQ (Single Correct Answer)

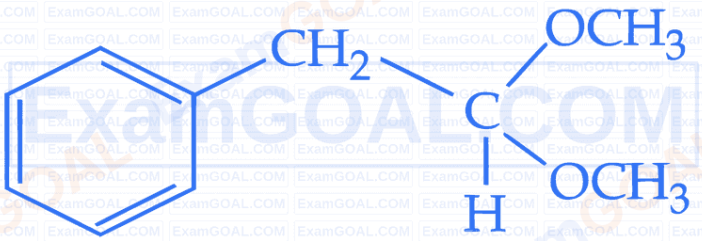
Open solution

Which one of the following, with HBr will give a phenol?

(A)



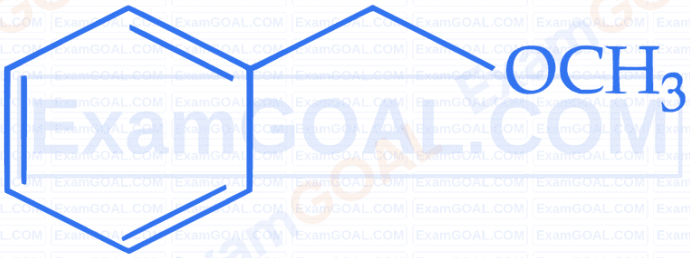
(B)



(C)



(D)



Q11. MCQ (Single Correct Answer)

Open solution

The calculated spin-only magnetic moments of $K_3[Fe(OH)_6]$ and $K_4[Fe(OH)_6]$ respectively are :

(A) 4.90 and 4.90 B.M.

(B) 4.90 and 5.92 B.M.

(C) 5.92 and 4.90 B.M.

(D) 3.87 and 4.90 B.M.

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Q12. MCQ (Single Correct Answer)

Open solution

Consider the equilibrium

$$CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g)$$

If the pressure applied over the system increases by two fold at constant temperature then

(A) Concentration of reactants and products increases.

(B) Equilibrium will shift in forward direction.

(C) Equilibrium constant increases since concentration of products increases.

(D) Equilibrium constant remains unchanged as concentration of reactants and products remain same.

Choose the **correct** answer from the options given below :

(A) (A) and (B) only

(B) (A), (B) and (D) only

(C) (B) and (C) only

(D) (A), (B) and (C) only

Q13. MCQ (Single Correct Answer)

Open solution

First ionisation enthalpy values of first four group 15 elements are given below. Choose the correct value for the element that is a main component of apatite family :

(A) 834 kJ mol^{-1}

(B) 947 kJ mol^{-1}

(C) 1402 kJ mol^{-1}

(D) 1012 kJ mol^{-1}

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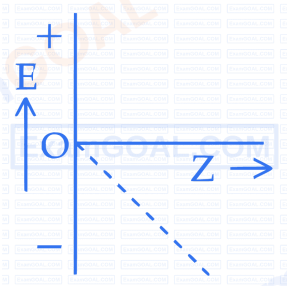
Q14. MCQ (Single Correct Answer)

Open solution

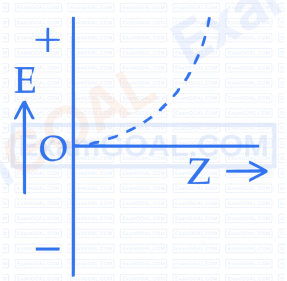
For hydrogen like species, which of the following graphs provides the most appropriate representation of E vs Z plot for a constant n?

[E : Energy of the stationary state, Z : atomic number, n = principal quantum number]

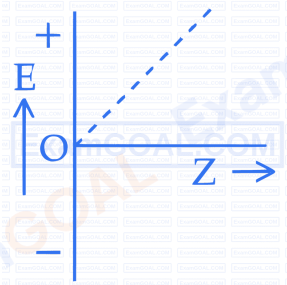
(A)



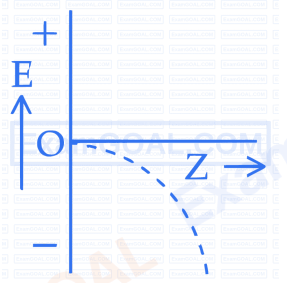
(B)



(C)



(D)



Q15. MCQ (Single Correct Answer)

Open solution

If C (diamond) $\rightarrow$ C (graphite) $+X\ \mathrm{kJ\ mol^{-1}}$

C (diamond) $+\mathrm{O_2(g)} \rightarrow \mathrm{CO_2(g)} + \mathrm{YkJmol^{-1}}$

C (graphite) $+\mathrm{O_2(g)} \rightarrow \mathrm{CO_2(g)} + \mathrm{ZkJmol^{-1}}$

at constant temperature. Then

(A) −X = Y + Z (B) X = Y − Z (C) X = −Y + Z (D) X = Y + Z

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Q16. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement (I): It is impossible to specify simultaneously with arbitrary precision, both the linear momentum and the position of a particle.

Statement (II) : If the uncertainty in the measurement of position and uncertainty in measurement of momentum are equal for an electron, then the uncertainty in the measurement of velocity is $\geqslant \sqrt{\frac{h}{\pi}} \times \frac{1}{2m}$.

In the light of the above statements, choose the correct answer from the options given below :

(A) **Statement I** is false but **Statement II** is true

(B) Both **Statement I** and **Statement II** are true

(C) Both **Statement I** and **Statement II** are false

(D) **Statement I** is true but **Statement II** is false

Q17. MCQ (Single Correct Answer)

Open solution

Total number of sigma (σ) _____ and pi(π) _____ bonds respectively present in hex-1-en-4-yne are:

(A) 14 and 3 (B) 11 and 3 (C) 13 and 3 (D) 3 and 13

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Q18. MCQ (Single Correct Answer)

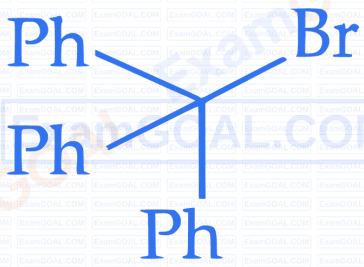
Which among the following halides will generate the most stable carbocation in the nucleophilic substitution reaction?

Open solution

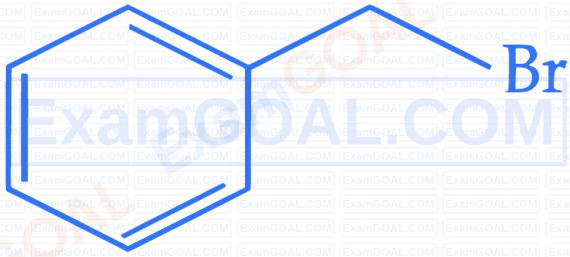
(A)



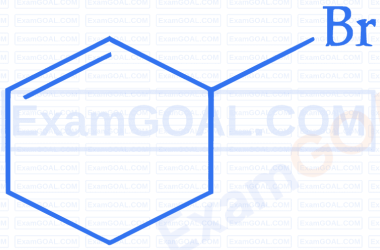
(B)



(C)



(D)



Q19. MCQ (Single Correct Answer)

Drug X becomes ineffective after 50% decomposition. The original concentration of drug in a bottle was 16mg/mL which becomes 4mg/mL in 12 months. The expiry time of the drug in months is _____.

Open solution

Assume that the decomposition of the drug follows first order kinetics.

(A) 12 (B) 3 (C) 6 (D) 2

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Q20. MCQ (Single Correct Answer)

O_2 gas will be evolved as a product of electrolysis of :

(A) an aqueous solution of $AgNO_3$ using silver electrodes.

(B) an aqueous solution of $AgNO_3$ using platinum electrodes.

(C) a dilute solution of H_2SO_4 using platinum electrodes.

(D) a high concentration solution of H_2SO_4 using platinum electrodes.

Choose the correct answer from the options given below :

(A) (B) and (C) only **(B)** (B) and (D) only **(C)** (A) and (D) only **(D)** (A) and (C) only

Open solution

Q21. Numerical

Total number of non-bonded electrons present in NO_2^- ion based on Lewis theory is _____.

Open solution

Q22. Numerical

Consider the following low-spin complexes

$K_3[Co(NO_2)_6], K_4[Fe(CN)_6], K_3[Fe(CN)_6], Cu_2[Fe(CN)_6]$ and $Zn_2[Fe(CN)_6]$

The sum of the spin-only magnetic moment values of complexes having yellow colour is _____ B.M. (answer in nearest integer)

Open solution

Q23. Numerical

In the sulphur estimation, 0.20 g of a pure organic compound gave 0.40 g of barium sulphate. The percentage of sulphur in the compound is _____ $\times 10^{-1}\%$.

(Molar mass : $O = 16,\ S = 32, Ba = 137$ in $g\,mol^{-1}$)

Open solution

Q24. Numerical

In the Claisen-Schmidt reaction to prepare dibenzalacetone from 5.3 g of benzaldehyde, a total of 3.51 g of product was obtained. The percentage yield in this reaction was _____ %.

Open solution

Q25. Numerical

Isomeric hydrocarbons → negative Baeyer's test

(Molecular formula C_9H_{12})

The total number of isomers from above with four different non-aliphatic substitution sites is -

Open solution

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

Let $S = \mathbf{N} \cup \{0\}$. Define a relation R from S to $\mathbf{R}$ by :

$R = \{(x, y) : \log_e y = x \log_e \left(\frac{2}{5}\right), x \in S, y \in \mathbf{R}\}$.

Then, the sum of all the elements in the range of R is equal to :

(A) $\frac{3}{2}$ (B) $\frac{10}{9}$ (C) $\frac{5}{2}$ (D) $\frac{5}{3}$

Open solution

Q27. MCQ (Single Correct Answer)

Let $A = [a_{ij}]$ be a 2×2 matrix such that $a_{ij} \in \{0, 1\}$ for all i and j. Let the random variable X denote the possible values of the determinant of the matrix A. Then, the variance of X is:

(A) $\frac{5}{8}$ (B) $\frac{1}{4}$ (C) $\frac{3}{4}$ (D) $\frac{3}{8}$

Open solution

Q28. MCQ (Single Correct Answer)

Let a straight line L pass through the point $P(2, -1, 3)$ and be perpendicular to the lines $\frac{x-1}{2} = \frac{y+1}{1} = \frac{z-3}{-2}$ and $\frac{x-3}{1} = \frac{y-2}{3} = \frac{z+2}{4}$. If the line L intersects the yz-plane at the point Q, then the distance between the points P and Q is:

(A) $\sqrt{10}$ (B) 2 (C) $2\sqrt{3}$ (D) 3

Open solution

Q29. MCQ (Single Correct Answer)

If the domain of the function $\log_5(18x - x^2 - 77)$ is (α, β) and the domain of the function $\log_{(x-1)}\left(\frac{2x^2+3x-2}{x^2-3x-4}\right)$ is (γ, δ), then $\alpha^2 + \beta^2 + \gamma^2$ is equal to:

(A) 186 (B) 179 (C) 195 (D) 174

Open solution

Q30. MCQ (Single Correct Answer)

Let $\hat{a}$ be a unit vector perpendicular to the vectors $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{c} = 2\hat{i} + 3\hat{j} - \hat{k}$, and $\hat{a}$ makes an angle of $\cos^{-1}\left(-\frac{1}{3}\right)$ with the vector $\hat{i} + \hat{j} + \hat{k}$. If $\hat{a}$ makes an angle of $\frac{\pi}{3}$ with the vector $\hat{i} + \alpha\hat{j} + \hat{k}$, then the value of α is:

(A) $\sqrt{3}$ (B) $\sqrt{6}$ (C) $-\sqrt{6}$ (D) $-\sqrt{3}$

Open solution

Q31. MCQ (Single Correct Answer)

Let $\alpha, \beta\ (\alpha \neq \beta)$ be the values of m, for which the equations $x + y + z = 1$, $x + 2y + 4z = m$ and $x + 4y + 10z = m^2$ have infinitely many solutions. Then the value of $\sum_{n=1}^{10}(n^\alpha + n^\beta)$ is equal to :

(A) 3410 (B) 560 (C) 3080 (D) 440

Open solution

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Q32. MCQ (Single Correct Answer)

If the set of all $a \in \mathbf{R}$, for which the equation $2x^2 + (a-5)x + 15 = 3a$ has no real root, is the interval (α, β), and $X = \{x \in Z; \alpha < x < \beta\}$, then $\sum_{x \in X} x^2$ is equal to:

(A) 2139 (B) 2119 (C) 2109 (D) 2129

Open solution

Q33. MCQ (Single Correct Answer)

Let the line *x + y = 1* meet the axes of *x* and *y* at *A* and *B*, respectively. A right angled triangle *AMN* is inscribed in the triangle *OAB*, where *O* is the origin and the points *M* and *N* lie on the lines *OB* and *AB*, respectively. If the area of the triangle *AMN* is $\frac{4}{9}$ of the area of the triangle *OAB* and *AN : NB* = $\lambda : 1$, then the sum of all possible value(s) of λ is:

(A) $\frac{1}{2}$ (B) $\frac{5}{2}$ (C) 2 (D) $\frac{13}{6}$

Open solution

Q34. MCQ (Single Correct Answer)

Bag 1 contains 4 white balls and 5 black balls, and Bag 2 contains *n* white balls and 3 black balls. One ball is drawn randomly from Bag 1 and transferred to Bag 2. A ball is then drawn randomly from Bag 2. If the probability, that the ball drawn is white, is $\frac{29}{45}$, then *n* is equal to:

(A) 5 (B) 6 (C) 4 (D) 3

Open solution

Q35. MCQ (Single Correct Answer)

The remainder, when 7^{103} is divided by 23, is equal to:

(A) 9 (B) 6 (C) 14 (D) 17

Open solution

Q36. MCQ (Single Correct Answer)

If $\alpha x + \beta y = 109$ is the equation of the chord of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$, whose mid point is $\left(\frac{5}{2}, \frac{1}{2}\right)$. then $\alpha + \beta$ is equal to :

(A) 37 (B) 46 (C) 72 (D) 58

Open solution

Q37. MCQ (Single Correct Answer)

Let a circle C pass through the points (4, 2) and (0, 2), and its centre lie on 3x + 2y + 2 = 0. Then the length of the chord, of the circle C, whose mid-point is (1, 2), is:

(A) $4\sqrt{2}$ (B) $2\sqrt{2}$ (C) $2\sqrt{3}$ (D) $\sqrt{3}$

Open solution

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Q38. MCQ (Single Correct Answer)

If for the solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx} + (\tan x)y = \frac{2+\sec x}{(1+2\sec x)^2}$, $x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$, $f\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{10}$, then $f\left(\frac{\pi}{4}\right)$ is equal to:

(A) $\frac{5-\sqrt{3}}{2\sqrt{2}}$ (B) $\frac{4-\sqrt{2}}{14}$

(C) $\frac{9\sqrt{3}+3}{10(4+\sqrt{3})}$ (D) $\frac{\sqrt{3}+1}{10(4+\sqrt{3})}$

Open solution

Q39. MCQ (Single Correct Answer)

Let $\mathrm{A} = [a_{ij}]$ be a matrix of order 3×3, with $a_{ij} = (\sqrt{2})^{i+j}$. If the sum of all the elements in the third row of A^2 is $\alpha + \beta\sqrt{2}, \alpha, \beta \in \mathbf{Z}$, then $\alpha + \beta$ is equal to :

(A) 210 (B) 280 (C) 224 (D) 168

Open solution

Q40. MCQ (Single Correct Answer)

If $\sin x + \sin^2 x = 1, x \in \left(0, \frac{\pi}{2}\right)$, then

$(\cos^{12} x + \tan^{12} x) + 3(\cos^{10} x + \tan^{10} x + \cos^8 x + \tan^8 x) + (\cos^6 x + \tan^6 x)$ is equal to:

(A) 3 (B) 4 (C) 2 (D) 1

Open solution

Q41. MCQ (Single Correct Answer)

Let $f(x) = \int_0^x \mathrm{t}\left(\mathrm{t}^2 - 9\mathrm{t} + 20\right)\mathrm{dt}, 1 \le x \le 5$. If the range of f is $[\alpha, \beta]$, then $4(\alpha + \beta)$ equals :

(A) 253 (B) 157 (C) 154 (D) 125

Open solution

Q42. MCQ (Single Correct Answer)

Let the area enclosed between the curves $|y| = 1 - x^2$ and $x^2 + y^2 = 1$ be α. If $9\alpha = \beta\pi + \gamma; \beta, \gamma$ are integers, then the value of $|\beta - \gamma|$ equals:

(A) 15 (B) 18 (C) 33 (D) 27

Open solution

Q43. MCQ (Single Correct Answer)

Let P be the foot of the perpendicular from the point $(1, 2, 2)$ on the line $\mathrm{L} : \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$. Let the line $\vec{r} = (-\hat{i} + \hat{j} - 2\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k}), \lambda \in \mathbf{R}$, intersect the line L at Q . Then $2(\mathrm{PQ})^2$ is equal to :

(A) 25 (B) 27 (C) 19 (D) 29

Open solution

Q44. MCQ (Single Correct Answer)

If all the words with or without meaning made using all the letters of the word "KANPUR" are arranged as in a dictionary, then the word at 440th position in this arrangement is :

(A) PRNAKU (B) PRKAUN (C) PRKANU (D) PRNAUK

Open solution

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Q45. MCQ (Single Correct Answer)

Let the function $f(x) = (x^2 - 1)\left|x^2 - ax + 2\right| + \cos|x|$ be not differentiable at the two points $x = \alpha = 2$ and $x = \beta$. Then the distance of the point (α, β) from the line $12x + 5y + 10 = 0$ is equal to :

Open solution

(A) 5 (B) 2 (C) 4 (D) 3

Q46. Numerical

If $24\int_0^{\frac{\pi}{4}} \left(\sin\left|4x - \frac{\pi}{12}\right| + [2\sin x]\right)dx = 2\pi + \alpha$, where $[\cdot]$ denotes the greatest integer function, then α is equal to _____.

Open solution

Q47. Numerical

Let $y^2 = 12x$ be the parabola and S be its focus. Let PQ be a focal chord of the parabola such that $(SP)(SQ) = \frac{147}{4}$. Let C be the circle described taking PQ as a diameter. If the equation of a circle C is $64x^2 + 64y^2 - \alpha x - 64\sqrt{3}y = \beta$, then $\beta - \alpha$ is equal to .

Open solution

Q48. Numerical

Let integers $a, b \in [-3, 3]$ be such that $a + b \neq 0$. Then the number of all possible ordered pairs (a, b), for which $\left|\frac{z-a}{z+b}\right| = 1$ and $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 1, z \in C$, where ω and ω^2 are the roots of $x^2 + x + 1 = 0$, is equal to _____ .

Open solution

Q49. Numerical

If $\lim\limits_{t\to 0}\left(\int_0^1 (3x+5)^t dx\right)^{\frac{1}{t}} = \frac{\alpha}{5e}\left(\frac{8}{5}\right)^{\frac{2}{3}}$, then α is equal to _____.

Open solution

Q50. Numerical

Let $a_1, a_2, \ldots, a_{2024}$ be an Arithmetic Progression such that $a_1 + (a_5 + a_{10} + a_{15} + \ldots + a_{2020}) + a_{2024} = 2233$. Then $a_1 + a_2 + a_3 + \ldots + a_{2024}$ is equal to _____.

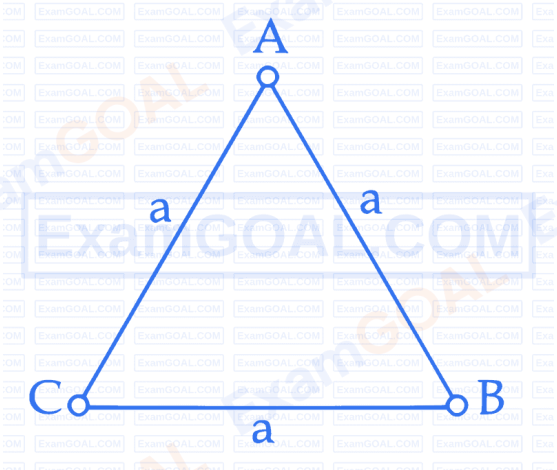
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Physics 25 questions

Q51. MCQ (Single Correct Answer)

Open solution



Three equal masses m are kept at vertices (A, B, C) of an equilateral triangle of side a in free space. At $t = 0$, they are given an initial velocity $\overrightarrow{V_A} = V_0\overrightarrow{AC}, \overrightarrow{V_B} = V_0\overrightarrow{BA}$ and $\overrightarrow{V_C} = V_0\overrightarrow{CB}$. Here, $\overrightarrow{AC}, \overrightarrow{CB}$ and $\overrightarrow{BA}$ are unit vectors along the edges of the triangle. If the three masses interact gravitationally, then the magnitude of the net angular momentum of the system at the point of collision is :

(A) $\frac{1}{2}$ a mV_0

(B) 3 a mV_0

(C) $\frac{3}{2}$ a mV_0

(D) $\frac{\sqrt{3}}{2}$ a mV_0

Q52. MCQ (Single Correct Answer)

Open solution

In an experiment with photoelectric effect, the stopping potential,

(A) is $\left(\frac{1}{e}\right)$ times the maximum kinetic energy of the emitted photoelectrons

(B) increases with increase in the intensity of the incident light

(C) decreases with increase in the intensity of the incident light

(D) increases with increase in the wavelength of the incident light

Q53. MCQ (Single Correct Answer)

Open solution

The number of spectral lines emitted by atomic hydrogen that is in the 4th energy level, is

(A) 3 (B) 6 (C) 1 (D) 0

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Q54. MCQ (Single Correct Answer)

The difference of temperature in a material can convert heat energy into electrical energy. To harvest the heat energy, the material should have

(A) low thermal conductivity and high electrical conductivity

(B) low thermal conductivity and low electrical conductivity

(C) high thermal conductivity and high electrical conductivity

(D) high thermal conductivity and low electrical conductivity

Open solution

Q55. MCQ (Single Correct Answer)

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : With the increase in the pressure of an ideal gas, the volume falls off more rapidly in an isothermal process in comparison to the adiabatic process.

Reason (R) : In isothermal process, PV = constant, while in adiabatic process PV^{γ} = constant. Here γ is the ratio of specific heats, P is the pressure and V is the volume of the ideal gas.

In the light of the above statements, choose the **correct** answer from the options given below:

(A) (A) is true but (R) is false

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(C) (A) is false but (R) is true

(D) Both (A) and (R) are true and (R) is the correct explanation of (A)

Open solution

Q56. MCQ (Single Correct Answer)

A sand dropper drops sand of mass m(t) on a conveyer belt at a rate proportional to the square root of speed (v) of the belt, i.e., $\frac{dm}{dt} \propto \sqrt{v}$. If P is the power delivered to run the belt at constant speed then which of the following relationship is true?

(A) P $\propto \sqrt{v}$ (B) P $\propto v$ (C) $P^2 \propto v^3$ (D) $P^2 \propto v^5$

Open solution

Q57. MCQ (Single Correct Answer)

A plane electromagnetic wave propagates along the + x direction in free space. The components of the electric field, $\vec{E}$ and magnetic field, $\vec{B}$ vectors associated with the wave in Cartesian frame are

(A) E_x, B_y (B) E_y, B_x (C) E_y, B_z (D) E_z, B_y

Open solution

Q58. MCQ (Single Correct Answer)

A cup of coffee cools from 90°C to 80°C in t minutes when the room temperature is 20°C. The time taken by the similar cup of coffee to cool from 80°C to 60°C at the same room temperature is:

(A) $\frac{13}{5}t$ (B) $\frac{10}{13}t$ (C) $\frac{5}{13}t$ (D) $\frac{13}{10}t$

Open solution

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Q59. MCQ (Single Correct Answer)

A convex lens made of glass (refractive index = 1.5) has focal length 24 cm in air. When it is totally immersed in water (refractive index = 1.33), its focal length changes to

(A) 96 cm (B) 72 cm (C) 24 cm (D) 48 cm

Open solution

Q60. MCQ (Single Correct Answer)

Two bodies A and B of equal mass are suspended from two massless springs of spring constant k_1 and k_2, respectively. If the bodies oscillate vertically such that their amplitudes are equal, the ratio of the maximum velocity of A to the maximum velocity of B is

(A) $\sqrt{\frac{k_2}{k_1}}$ (B) $\sqrt{\frac{k_1}{k_2}}$ (C) $\frac{k_2}{k_1}$ (D) $\frac{k_1}{k_2}$

Open solution

Q61. MCQ (Single Correct Answer)

Match **List - I** with **List - II**.

List - I	List - II
(A) Young's Modulus	(I) $M L^{-1} T^{-1}$
(B) Torque	(II) $M L^{-1} T^{-2}$
(C) Coefficient of Viscosity	(III) $M^{-1} L^{3} T^{-2}$
(D) Gravitational Constant	(IV) $M L^{2} T^{-2}$

Choose the **correct** answer from the options given below :

(A) (A)-(I), (B)-(III), (C)-(II), (D)-(IV) **(B)** (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

(C) (A)-(II), (B)-(I), (C)-(IV), (D)-(III) **(D)** (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Open solution

Q62. MCQ (Single Correct Answer)

Match List - I with List - II.

List - I	List - II
(A) Magnetic induction	(I) Ampere meter^2
(B) Magnetic intensity	(II) Weber
(C) Magnetic flux	(III) Gauss
(D) Magnetic moment	(IV) Ampere meter^{-1}

Choose the correct answer from the options given below:

(A) (A)-(III), (B)-(IV), (C)-(II), (D)-(I) **(B)** (A)-(III), (B)-(II), (C)-(I), (D)-(IV)

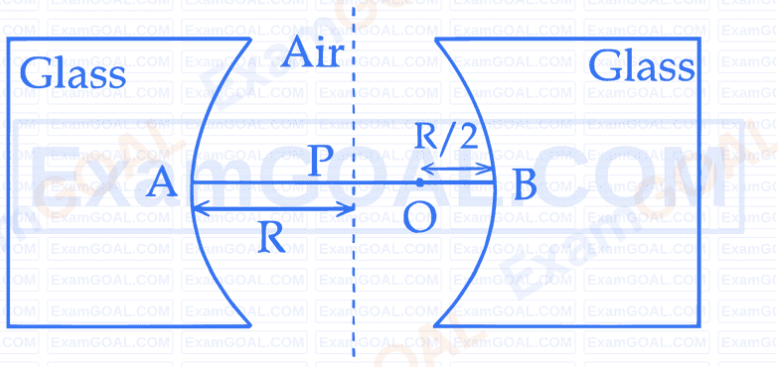
(C) (A)-(III), (B)-(IV), (C)-(I), (D)-(II) **(D)** (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

Open solution

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Q63. MCQ (Single Correct Answer)

Open solution



Two concave refracting surfaces of equal radii of curvature and refractive index 1.5 face each other in air as shown in figure. A point object O is placed midway, between P and B. The separation between the images of O, formed by each refracting surface is :

(A) 0.124R (B) 0.114R (C) 0.411R (D) 0.214R

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Q64. MCQ (Single Correct Answer)

Open solution

The truth table for the circuit given below is:



(A)

A	B	Y
0	0	0
1	1	1
1	0	1
0	1	1

(B)

A	B	Y
0	0	0
1	0	1
0	1	0
1	1	0

(C)

A	B	Y
0	0	0
1	0	0
1	1	0
0	1	1

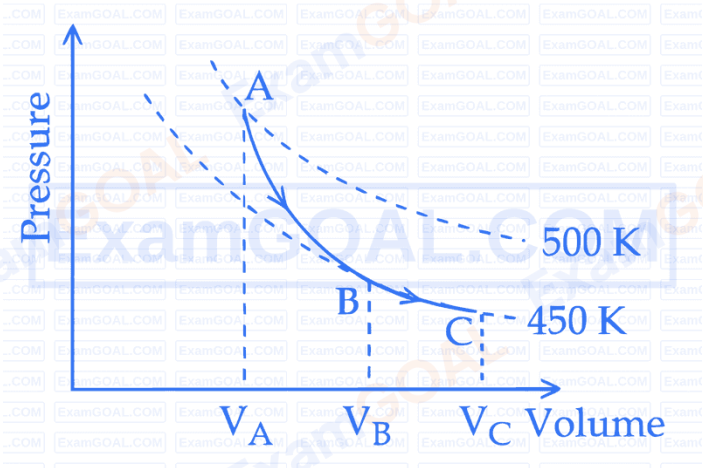
(D)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

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Q65. MCQ (Single Correct Answer)

Open solution



A poly-atomic molecule ($C_V = 3R, C_P = 4R$, where R is gas constant) goes from phase space point A $\left(\mathrm{P_A} = 10^5\ \mathrm{Pa},\ \mathrm{V_A} = 4 \times 10^{-6}\ \mathrm{m}^3\right)$ to point B $\left(\mathrm{P_B} = 5 \times 10^4\ \mathrm{Pa},\ \mathrm{V_B} = 6 \times 10^{-6}\ \mathrm{m}^3\right)$ to point C $\left(\mathrm{P_C} = 10^4\ \mathrm{Pa}, \mathrm{V}_C = 8 \times 10^{-6}\ \mathrm{m}^3\right)$. A to B is an adiabatic path and B to C is an isothermal path.

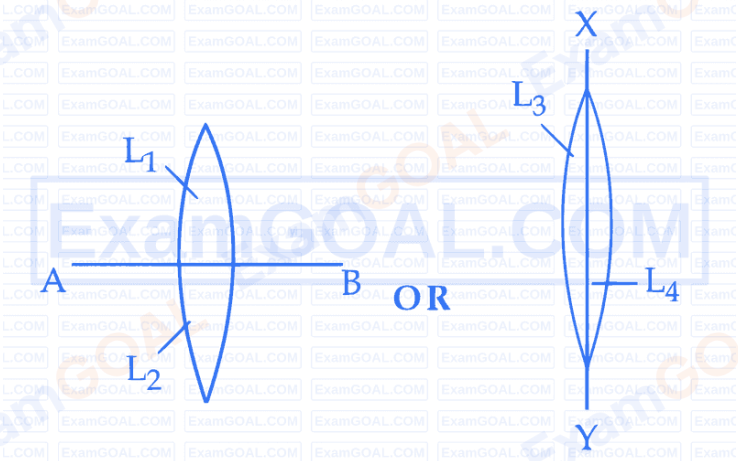
The net heat absorbed per unit mole by the system is :

(A) $500\mathrm{R}(\ln 3 + \ln 4)$ (B) $450\mathrm{R}(\ln 4 - \ln 3)$ (C) $500\mathrm{R}\ln 2$ (D) $400\mathrm{R}\ln 4$

Q66. MCQ (Single Correct Answer)

Open solution

Two identical symmetric double convex lenses of focal length f are cut into two equal parts L_1, L_2 by AB plane and L_3, L_4 by XY plane as shown in figure respectively. The ratio of focal lengths of lenses L_1 and L_3 is



(A) 1 : 2 (B) 1 : 1 (C) 2 : 1 (D) 1 : 4

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Q67. MCQ (Single Correct Answer)

Open solution

A point charge causes an electric flux of $-2 \times 10^4 \mathrm{Nm^2C^{-1}}$ to pass through a spherical Gaussian surface of 8.0 cm radius, centred on the charge. The value of the point charge is :

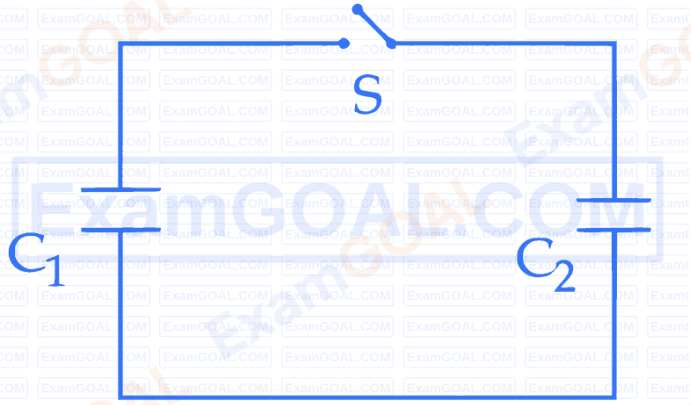
(Given $\epsilon_0 = 8.85 \times 10^{-12} \mathrm{C^2\,N^{-1}\,m^{-2}}$)

(A) $17.7 \times 10^{-8}\mathrm{C}$ (B) $-17.7 \times 10^{-8}\mathrm{C}$ (C) $15.7 \times 10^{-8}\mathrm{C}$ (D) $-15.7 \times 10^{-8}\mathrm{C}$

Q68. MCQ (Single Correct Answer)

Open solution

A capacitor, $C_1 = 6\mu F$ is charged to a potential difference of $V_0 = 5V$ using a 5V battery. The battery is removed and another capacitor, $C_2 = 12\mu F$ is inserted in place of the battery. When the switch 'S' is closed, the charge flows between the capacitors for some time until equilibrium condition is reached. What are the charges (q_1 and q_2) on the capacitors C_1 and C_2 when equilibrium condition is reached.



(A) $q_1 = 10\mu C,\ q_2 = 20\mu C$

(B) $q_1 = 15\mu C,\ q_2 = 30\mu C$

(C) $q_1 = 20\mu C,\ q_2 = 10\mu C$

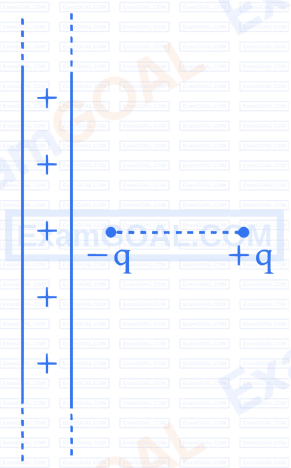
(D) $q_1 = 30\mu C,\ q_2 = 15\mu C$

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Q69. MCQ (Single Correct Answer)

Open solution

An electric dipole is placed at a distance of 2 cm from an infinite plane sheet having positive charge density σ_0. Choose the correct option from the following.



(A) Torque on dipole is zero and net force is directed away from the sheet.

(B) Potential energy of dipole is minimum and torque is zero.

(C) Potential energy and torque both are maximum.

(D) Torque on dipole is zero and net force acts towards the sheet.

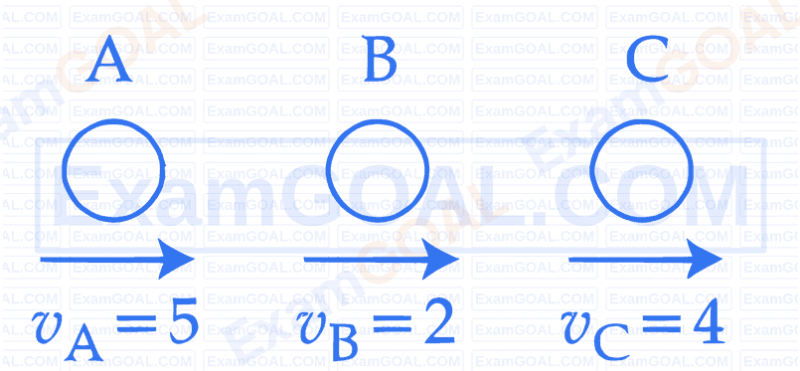
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Q70. MCQ (Single Correct Answer)

Open solution

Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A):



Three identical spheres of same mass undergo one dimensional motion as shown in figure with initial velocities $v_A = 5\,m/s, v_B = 2\,m/s, v_C = 4\,m/s$. If we wait sufficiently long for elastic collision to happen, then $v_A = 4\,m/s, v_B = 2\,m/s, v_C = 5\,m/s$ will be the final velocities.

Reason (R): In an elastic collision between identical masses, two objects exchange their velocities.

In the light of the above statements, choose the **correct** answer from the options given below:

(A) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

(C) (A) is false but (R) is true

(D) (A) is true but (R) is false

Q71. Numerical

Open solution

A physical quantity Q is related to four observables a, b, c, d as follows :

$Q = \frac{ab^4}{cd}$

where, $a = (60 \pm 3)Pa; b = (20 \pm 0.1)m; c = (40 \pm 0.2)Nsm^{-2}$ and $d = (50 \pm 0.1)m$, then the percentage error in Q is $\frac{x}{1000}$, where $x =$ _____ .

Q72. Numerical

Open solution

Two cars P and Q are moving on a road in the same direction. Acceleration of car P increases linearly with time whereas car Q moves with a constant acceleration. Both cars cross each other at time t = 0, for the first time. The maximum possible number of crossing(s) (including the crossing at t = 0) is _____.

Q73. Numerical

Open solution

Two planets, A and B are orbiting a common star in circular orbits of radii R_A and R_B, respectively, with $R_B = 2R_A$. The planet B is $4\sqrt{2}$ times more massive than planet A. The ratio $\left(\frac{L_B}{L_A}\right)$ of angular momentum (L_B) of planet B to that of planet A (L_A) is closest to integer _____.

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Q74. Numerical

A parallel plate capacitor consisting of two circular plates of radius 10 cm is being charged by a constant current of 0.15 A . If the rate of change of potential difference between the plates is $7 \times 10^8\ \mathrm{V/s}$ then the integer value of the distance between the parallel plates is

(Take, $\epsilon_0 = 9 \times 10^{-12} \frac{\mathrm{F}}{\mathrm{m}}, \pi = \frac{22}{7}$) _____ $\mu\mathrm{m}$.

Open solution

Q75. Numerical

The magnetic field inside a 200 turns solenoid of radius 10 cm is $2.9 \times 10^{-4}\ \mathrm{Tesla}$. If the solenoid carries a current of 0.29 A , then the length of the solenoid is _____ $\pi\mathrm{cm}$.

Open solution

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Answer Key

Chemistry

1. **B**	2. **A**	3. **C**	4. **A**	5. **D**
6. **D**	7. **C**	8. **C**	9. **A**	10. **A**
11. **C**	12. **A**	13. **D**	14. **D**	15. **B**
16. **B**	17. **C**	18. **B**	19. **C**	20. **A**
21. **12**	22. **0**	23. **275**	24. **60**	25. **2**

Mathematics

26. **D**	27. **D**	28. **D**	29. **A**	30. **C**
31. **D**	32. **A**	33. **C**	34. **B**	35. **C**
36. **D**	37. **C**	38. **B**	39. **C**	40. **C**
41. **B**	42. **C**	43. **B**	44. **B**	45. **D**
46. **12**	47. **1328**	48. **10**	49. **64**	50. **11132**

Physics

51. **D**	52. **A**	53. **B**	54. **A**	55. **D**
56. **D**	57. **C**	58. **A**	59. **A**	60. **B**
61. **D**	62. **A**	63. **B**	64. **D**	65. **B**
66. **A**	67. **B**	68. **A**	69. **B**	70. **C**
71. **7700**	72. **3**	73. **8**	74. **1320**	75. **8**

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